

Continuous growth or decay

Here $Q_0 > 0$ is the initial amount and $t \geq 0$ is elapsed time.

$$Q(t) = Q_0 e^{kt}$$

How does the sign of k distinguish growth from decay?

Changing to base e

Q is an amount in dollars and t is elapsed years, $t \geq 0$.

$$Q(t) = 250(1.12)^t$$

What equivalent base-e model uses an exact exponent coefficient?

Changing from base e

Q is mass in grams and t is elapsed hours, $t \geq 0$.

$$Q(t) = 600e^{-0.2t}$$

What equivalent model has a constant base raised to t?

Half-life

Here $M_0 > 0$ is initial mass, $t \geq 0$ is elapsed time, and $h > 0$ uses the same time unit.

$$M(t) = M_0 \left(\frac{1}{2} \right)^{t/h}$$

What does h measure?

Mass after several half-lives

A sample begins with 160 grams and has an 8-hour half-life. Assume exponential decay.

How many grams remain after 24 hours?

Finding a continuous decay rate

A sample begins with 100 grams and has a 6-hour half-life. Assume exponential decay and $t \geq 0$.

What exact base-e model gives the mass after t hours?

Doubling time

A population begins with 300 organisms and doubles every 5 days. Assume unrestricted exponential growth and $t \geq 0$.

What exponential model gives the count after t days?

Newton cooling model

T is temperature, T_0 is initial temperature, T_s is constant surrounding temperature, $k > 0$, and $t \geq 0$ is elapsed time.

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

Which quantity represents the surrounding temperature?

Interpreting a cooling curve

T is temperature in degrees Celsius and t is elapsed minutes, $t \geq 0$.

$$T(t) = 20 + 70e^{-0.1t}$$

What are the initial and long-term temperatures?

A cooling target

T is temperature in degrees Celsius and t is elapsed minutes, $t \geq 0$.

$$T(t) = 20 + 70e^{-0.1t}$$

When does the model reach 40 degrees Celsius, to the nearest tenth of a minute?

A cooling constant

An object cools from 80°C to 50°C in 10 minutes in a constant 20°C room. Use Newton's model with time in minutes.

What value of k fits the observation?

Logistic growth

Here $L > 0$, $A > 0$, $k > 0$, and $t \geq 0$ is elapsed time.

$$P(t) = \frac{L}{1 + Ae^{-kt}}$$

Which parameter is the upper limiting population?

A logistic population

P is a population count and t is elapsed weeks, $t \geq 0$.

$$P(t) = \frac{1200}{1 + 5e^{-0.4t}}$$

What are the initial and limiting populations?

A logistic threshold

P is a population count and t is elapsed weeks, $t \geq 0$.

$$P(t) = \frac{1200}{1 + 5e^{-0.4t}}$$

When does the model reach 600, to the nearest tenth of a week?

Choosing a model

Consider linear, exponential, logarithmic, and logistic models for elapsed time $t \geq 0$.

Which model family matches a positive initial population approaching a fixed upper limit?

Model limits

A positive population grows in a habitat with limited space and resources.

What feature makes an unrestricted exponential model questionable for long-term predictions?